



FPT UNIVERSITY

Campus Local Guide

Software Requirement Specification

– Hanoi, August 2022 –

Record of changes

Date	A* M, D	In charge	Change Description
15/May	A	DuongNX	Add Context Diagram and Introduction (I.1)
15/May	A	MinhTD	Add vendor use case diagram (I.2)

*A - Added M - Modified D - Deleted

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I. Overview

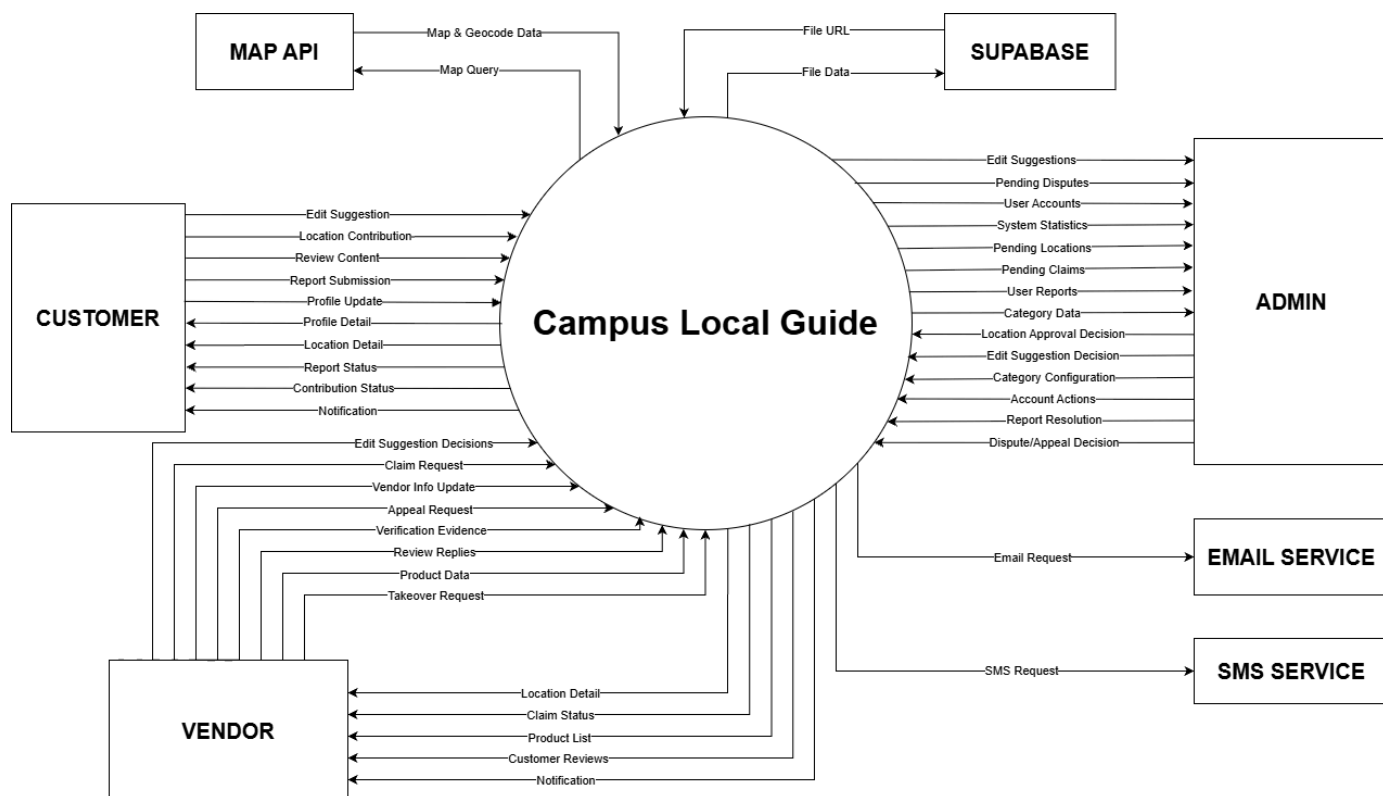
1. Introduction

The **Campus Local Guide** is a community-driven, all-in-one location discovery platform designed specifically for students living and studying in campus-adjacent areas. The system allows students to search, view, contribute, and review locations in their vicinity — including food & beverage establishments, rental accommodations, pharmacies, convenience stores, and local services — through both a map-based and list-based interface.

The System operates on a dual-contribution model:

- **Community-contributed content:** Students can submit new locations, write reviews, and report inaccurate information, collectively maintaining a living, up-to-date directory.
- **Vendor self-service:** Business owners can register or claim ownership of their locations, manage their business information, and respond to customer reviews.

All submitted content passes through an **Admin moderation queue** before being published.

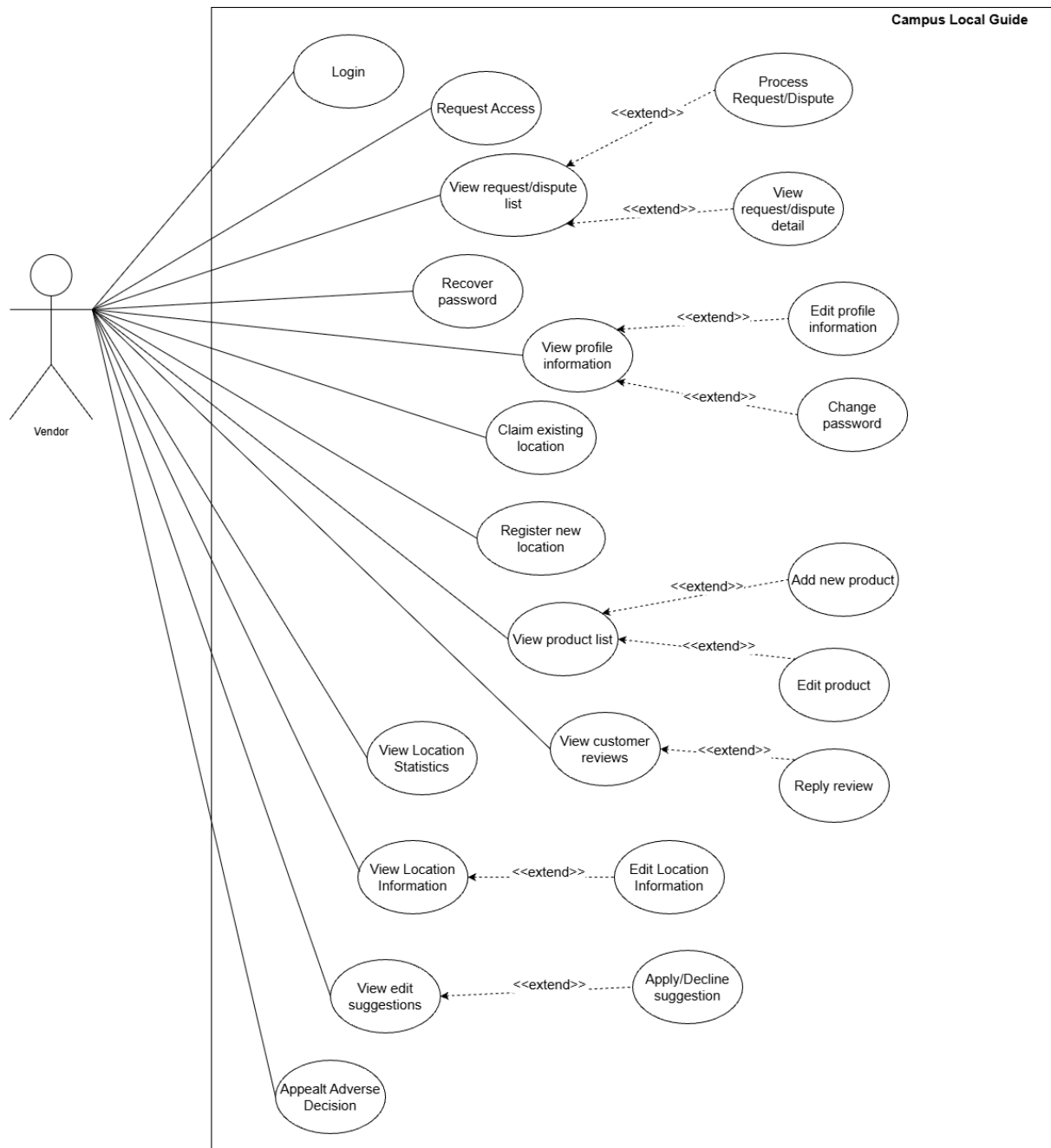


2. System Functions

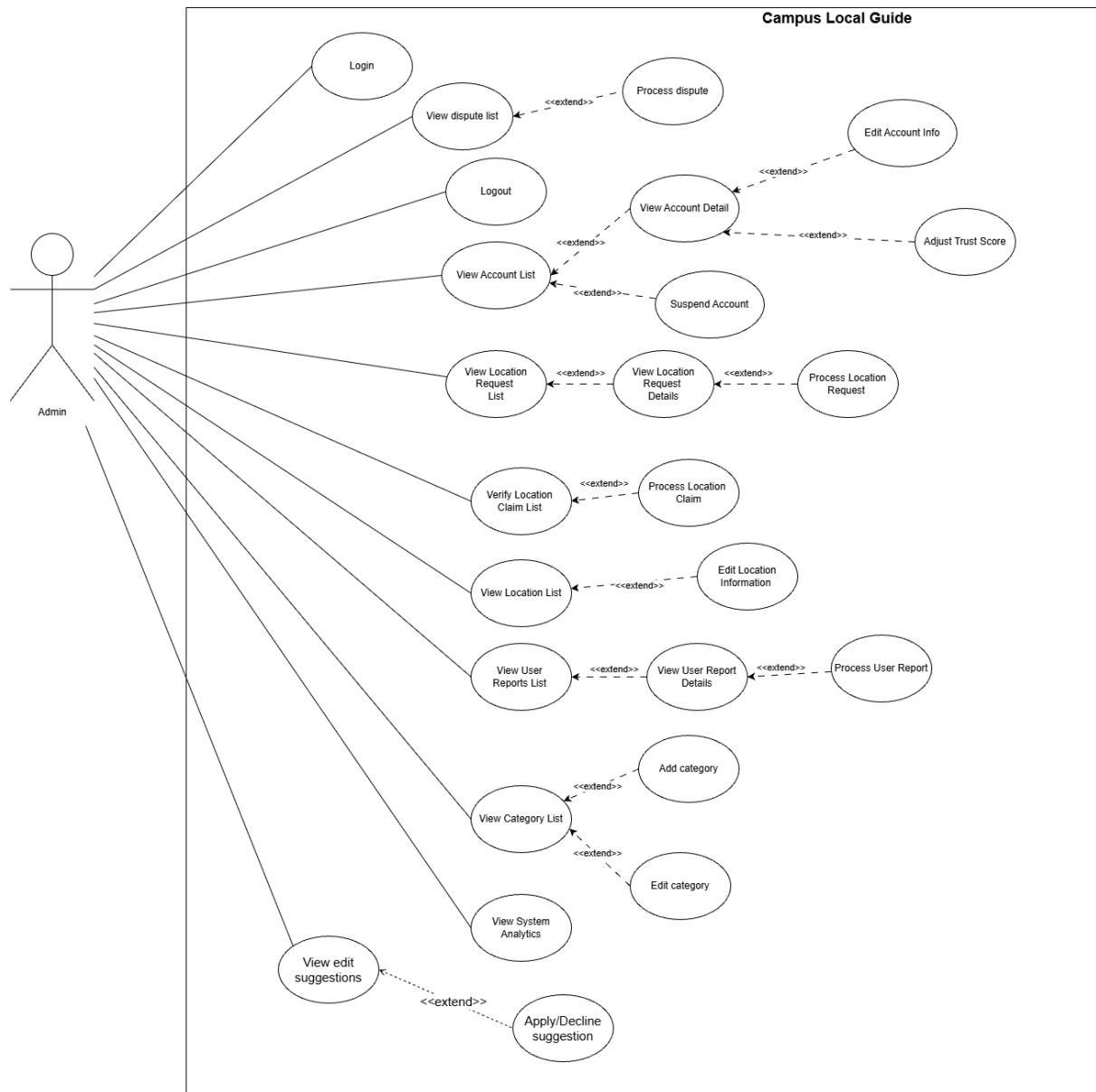
a. Use Case Diagram



Customer Use Case Diagram



Vendor Use Case Diagram



Admin Use Case Diagram

b. Screen Flow

[This part shows the system screens and the relationship among screens. You can draw the Screens Flow for the system in the form of diagram as below]

c. Screen Details

[Provide the descriptions for the screens in the Screens Flow above]

#	Feature	Screen	Description
1	Order Meals	Create Order	<<Screen Brief description>>
2	Order Meals	Change Order	
3	..		

d. User Authorization

Screen	User	Admin	Vendor
Authentication Screen	X	X	X
Login	X	X	X
Register	X		X
Forgot Password	X		X
Profile Screen	X	X	X
View Profile	X	X	X
Update Profile	X	X	X
Place Discovery Screens	X	X	X
Map	X	X	X
Get all published data	X	X	X
Location Detail	X	X	X
Bookmark Location	X		
Share Location Link	X	X	X
Report Location	X		
Contribution Screen	X		X
Submit New Location	X		X
Claim Location	X		
Manage Product			X
Update Product			X
Add New Product			X
Manage Place Information			X
Vendor Dashboard			X
Admin Dashboard		X	
Place Request List		X	
Place Request Detail		X	
Report List		X	
Report Detail		X	
Account List		X	
Update Account (Ban/Unban)		X	

In which:

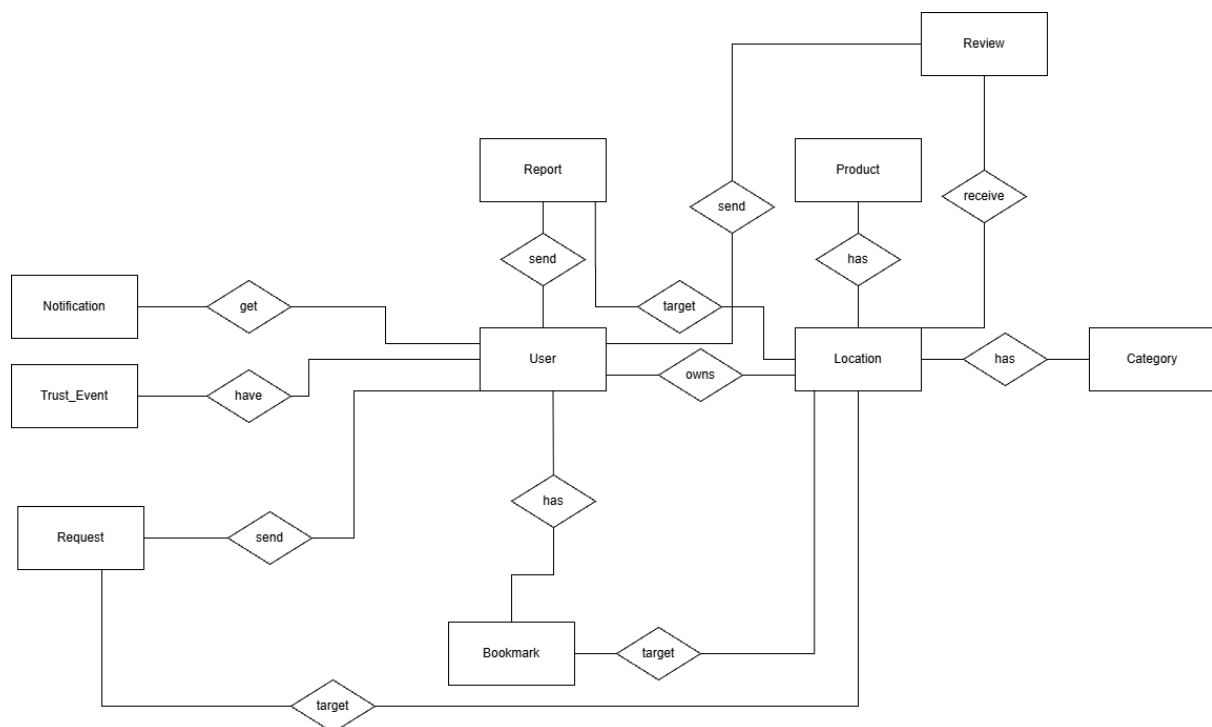
- Customer: The primary end-users of the platform, mainly students living near the campus. They act as both consumers of information and community contributors.
- Admin: The platform operators responsible for maintaining data integrity, quality control, and resolving disputes.
- Vendor: Local service providers, business owners, or landlords aiming to manage their brand presence and reach the student demographic.

e. Non-Screen Functions

[Provide the descriptions for the non-screen system functions, i.e batch/cron job, service, API, etc.]

#	Feature	System Function	Description
1	Authentication & User Account	Authentication API	Handles user registration, login, logout, password reset, session validation, and role-based access control for Customer, Vendor, and Admin.
2	Location Handling	Location Handling Service	Processes fused device location, accuracy radius, manual pin adjustment, reverse geocoding, and device-to-pin distance.
3	Geographic Control	Geographic Scoping Service	Checks whether submitted and displayed locations are within the allowed Hòa Lạc service radius.
4	Duplicate Prevention	Duplicate Detection Service	Detects suspected duplicate locations using name similarity and Haversine distance, then marks them for warning or Admin review.
5	Trust System	Trust Score Engine	Automatically updates user trust score based on approved submissions, valid reports, kept reviews, rejected content, and false reports
6	View Tracking	Location View Tracking Service	Records location views and reduces duplicate view counting.

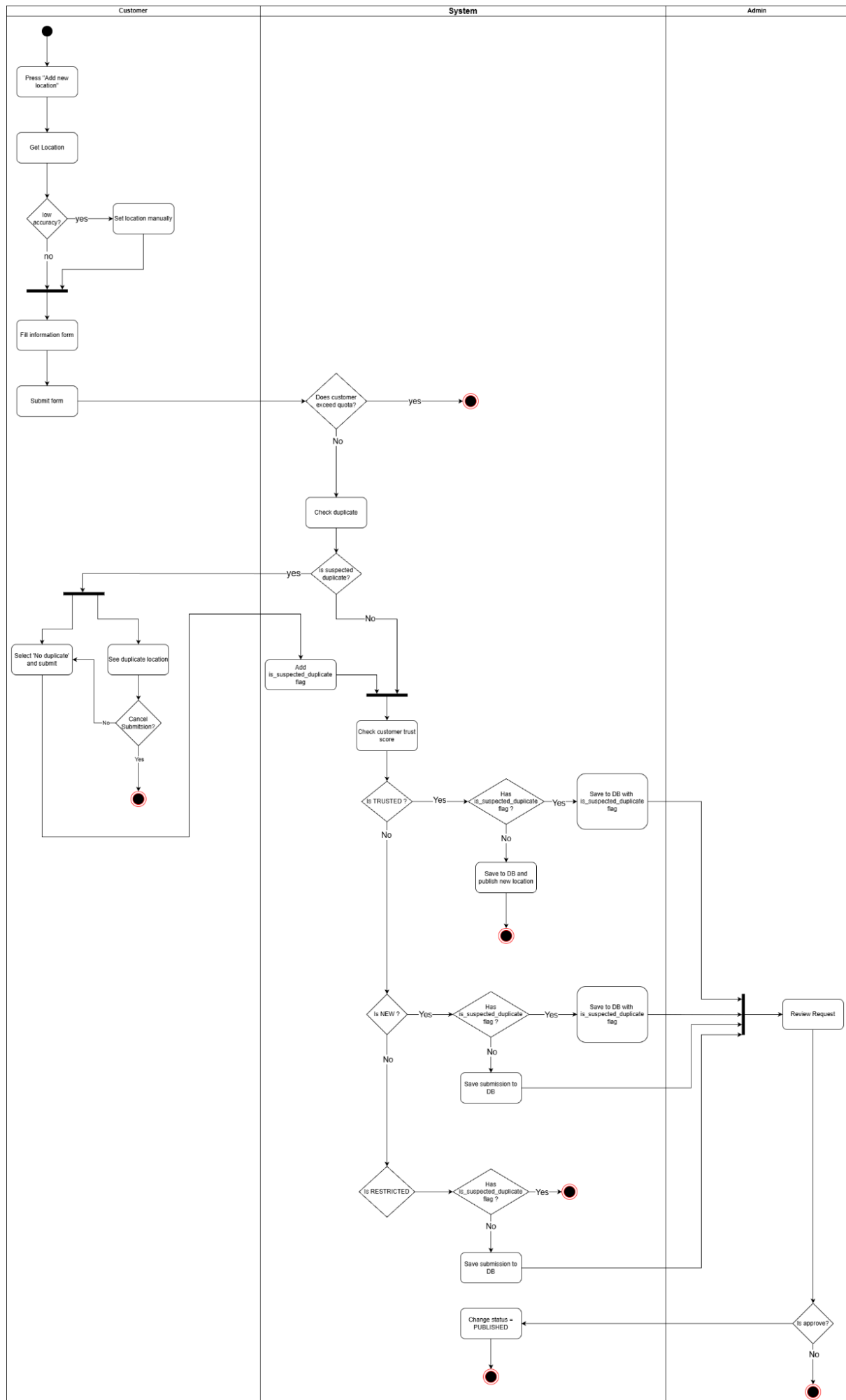
3. Entity Relationship Diagram



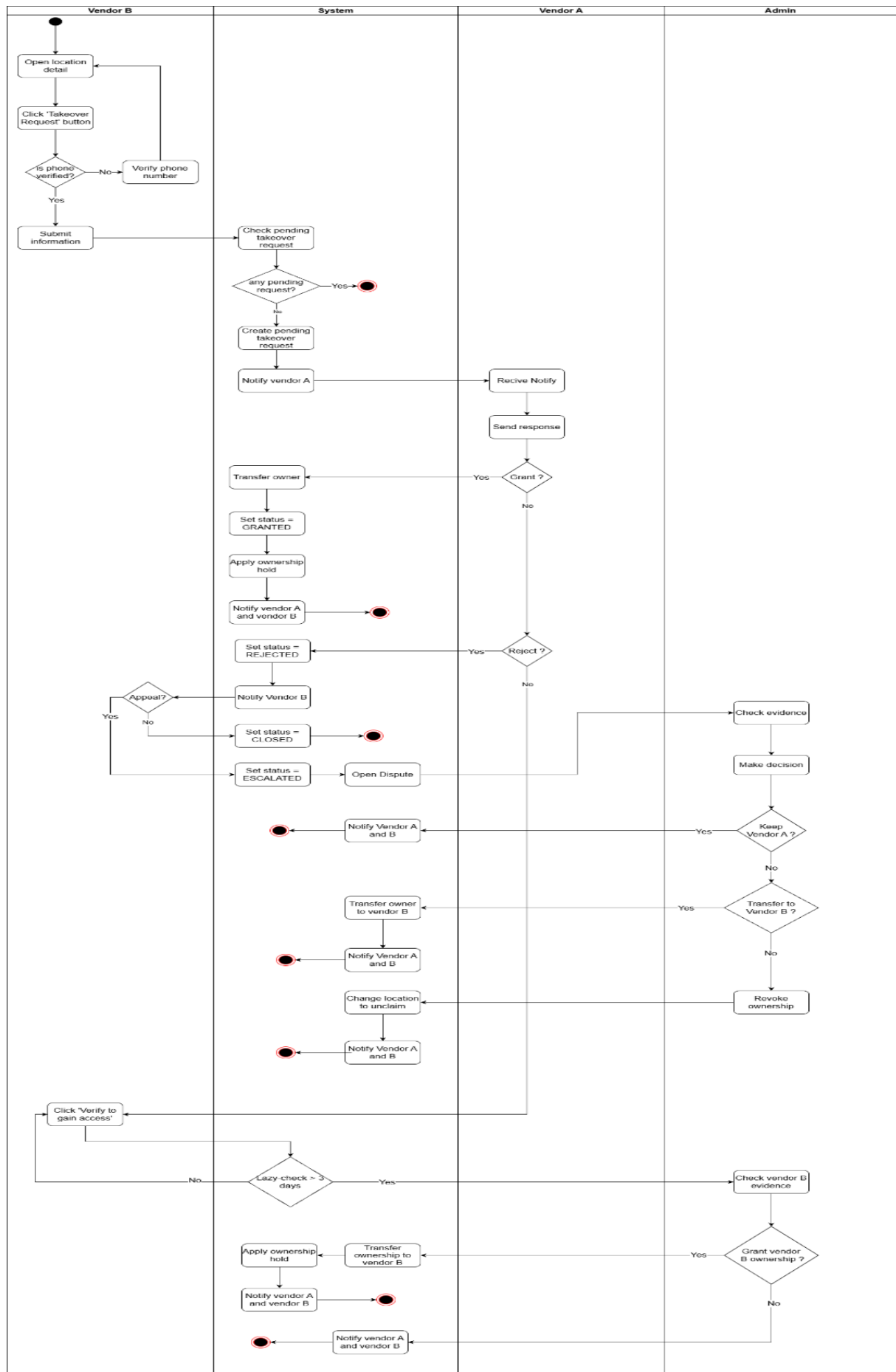
Entities Description

#	Entity	Description
1	User	Stores account information for all system actors, including Customers, Vendors, and Admins.
2	Location	Represents a physical place, service location, or establishment managed and displayed in the system.
3	Product	Represents items or services offered by a location, such as menu items, products, or service packages.
4	Notification	Handles system-generated alerts and push notifications sent to users.
5	Review	Stores customer reviews, ratings, comments, and feedback for a location.
3	Report	Logs user-generated reports about inaccurate information, inappropriate content, spam, closed places, or ownership issues.
7	Bookmark	A junction entity that tracks which users have saved which locations to their personal favorite list.
8	Category	Represents top-level classifications for locations, such as food, study, entertainment, or services.
9	Request	Stores user or vendor requests related to a location, such as creating, updating, claiming, or requesting access to manage a location.
10	Trust_Event	Stores trust score change records for users, including positive or negative events caused by approved submissions, valid reports, rejected content, or false reports.

4. Swimlane Diagram



Customer Contribute Location



II. Functional Requirements

1. <<Feature Name 1>>

a. <<Function Name 1>>

[A function can be a screen or a non-screen function (listed in the part I.2.d above). In this part, you need to provide the details on the related function, focus on mentioning below information

- Function trigger: how this function is triggered (navigation path, a timing frequency, etc.)*
- Function description: actors/roles, purpose, interface, data processing, etc.*
- Screen layout: mockup prototype of the screen, sample below is for Manage Products screen*
- Function Details: provide explanation for the data, validation, business logics, functionalities (for both normal cases and abnormal cases), etc. of the function so that the reader can image how it work.*

]

b. <<Function Name 2>>

...

2. <<Feature Name 2>>

...